

قناتنا على الواتساب

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Fractions

1. $\frac{a}{b} = a \div \frac{b}{1} = \frac{a}{1} \times \frac{1}{b} = \frac{a}{b}$

2. $\frac{a}{(b/c)} = a \div \frac{b}{c} = \frac{a}{1} \times \frac{c}{b} = \frac{ac}{b}$

3. $\frac{(a/b)}{c} = \frac{a}{b} \div \frac{c}{1} = \frac{a}{b} \times \frac{1}{c} = \frac{a}{bc}$

4. $\frac{(a/b)}{(c/d)} = \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$

Conversions

كتاب المفاهيم

P[Peta] $\times 10^{15}$
T[Tera] $\times 10^{12}$
G[Giga] $\times 10^{9}$
M[Mega] $\times 10^{6}$
K[Kilo] $\times 10^{3}$

Standard: meter, gram, second, Ampere, Unit

d[deci] $\times 10^{-1}$
c[centi] $\times 10^{-2}$
m[mili] $\times 10^{-3}$
μ[Micro] $\times 10^{-6}$
n[nano] $\times 10^{-9}$
p[pico] $\times 10^{-12}$
f[femto] $\times 10^{-15}$

A° [Angestrom] $\times 10^{-10}$ m

$Cm \xrightarrow{\times 10^{-2}} m$

$Cm^2 \xrightarrow{\times 10^{-4}} m^2$

$Cm^{-3} \xrightarrow{\times 10^{6}} m^{-3}$

$\mu m \xrightarrow{\times 10^{-3}} mm$

$\times 10^{[\text{من}] - [\text{الى}]}$

$\times 10^{[-6] - [-3]}$

Algebra

1. $I_1 \times I_1 = I_1^2$
2. $3I_1 + 4I_1 = 7I_1$
3. $[4R^2]^2 = 16R^4$
4. $\sqrt{9x_L^2} = \pm 3x_L$
5. $\frac{4L^2}{8L} = \frac{4}{8} \times \frac{L^2}{L} = 0.5L$
6. $m\sqrt{\frac{2KE}{m}} = \sqrt{m^2 \times \frac{2KE}{m}} = \sqrt{2mKE}$
7. $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$
8. $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$
9. $\frac{a}{b} = \frac{c}{d} \rightarrow b = \frac{ad}{c}$

OR $a \times d = b \times c$

Physics Universe

Mr. Abdelwahab

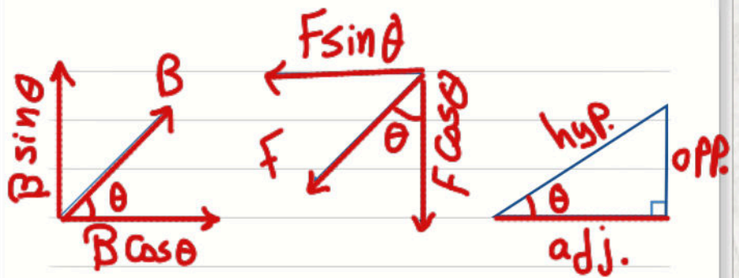
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Trignometry

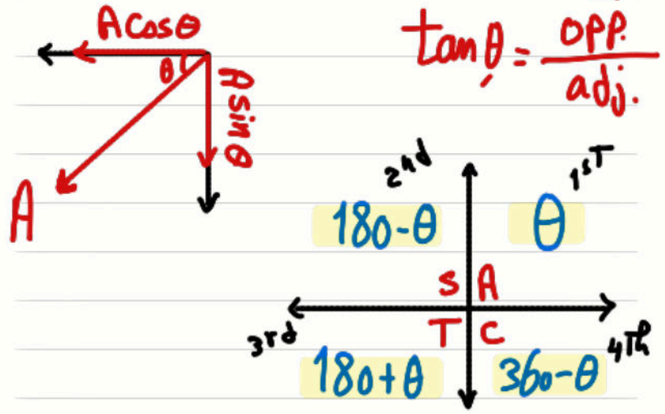
Resolution



sin البعيد ياخد $\sin\theta = \frac{opp.}{hyp.}$

cos القريب ياخد $\cos\theta = \frac{adj.}{hyp.}$

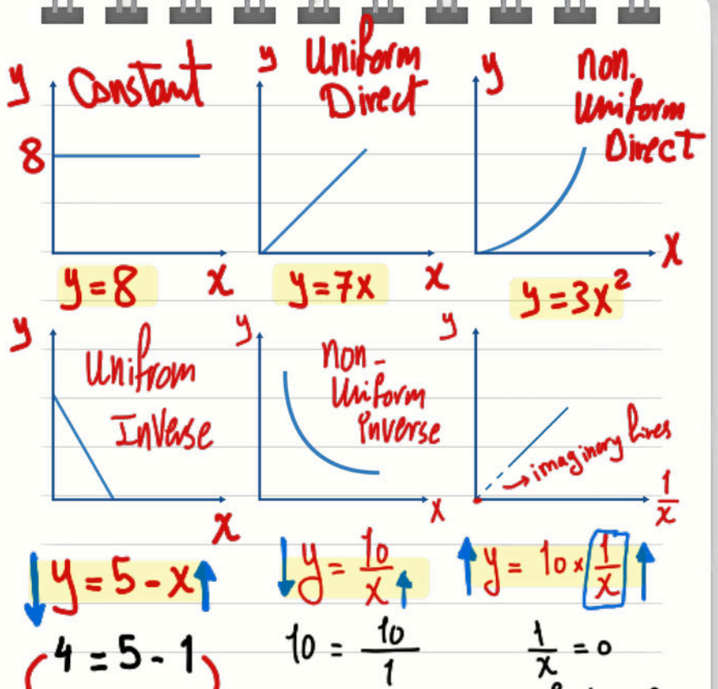
$\tan\theta = \frac{opp.}{adj.}$



Relations

constant · Direct · InVerse

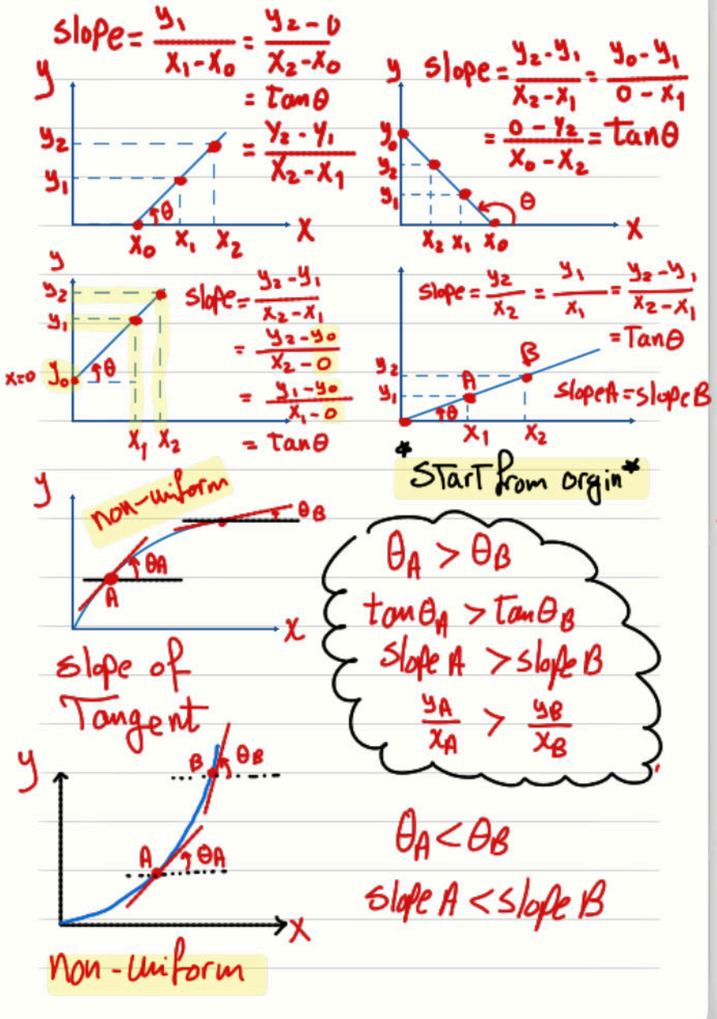
uniform → slope = const. non uniform → slope ≠ const.



Slope

الميل

$\text{Slope} = \frac{\Delta y}{\Delta x} = \tan\theta$



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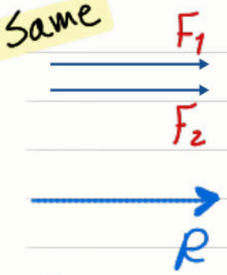
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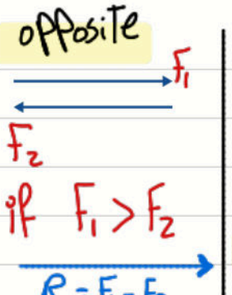
Resultant / Total / Net

Same



$R = F_1 + F_2$

opposite



if $F_1 > F_2$

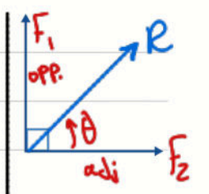
$R = F_1 - F_2$

if $F_1 < F_2$

$R = F_2 - F_1$

if $F_1 = F_2$

$R = Zero$



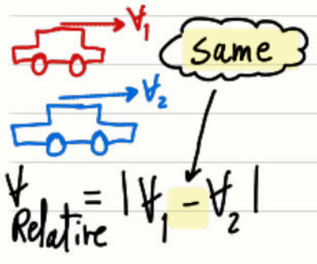
normal

perpendicular

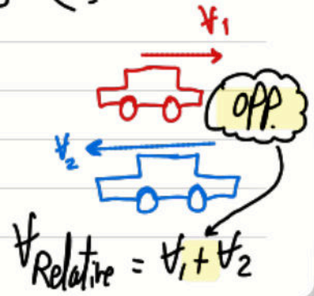
$R = \sqrt{F_1^2 + F_2^2}$

$\tan\theta = \frac{F_1}{F_2}$

* Relative velocity (السرعة النسبية)



$V_{Relative} = |V_1 - V_2|$



$V_{Relative} = V_1 + V_2$

Ratio

$M \overset{\text{direct}}{=} \frac{X\,y}{K\,L} \rightarrow \frac{M_1}{M_2} = \frac{x_1}{x_2} \times \frac{y_1}{y_2} \times \frac{K_2}{K_1} \times \frac{L_2}{L_1}$

Inverse

$R = \frac{\rho_e L}{A} \rightarrow \frac{R_1}{R_2} = \frac{\rho_{e1}}{\rho_{e2}} \times \frac{L_1}{L_2} \times \frac{A_2}{A_1}$

Const. Cancelld in any Ratio

$R = \frac{\rho_e L}{\pi r^2} \rightarrow \frac{R_1}{R_2} = \frac{\rho_{e1}}{\rho_{e2}} \times \frac{L_1}{L_2} \times \left[\frac{r_2}{r_1}\right]^2$

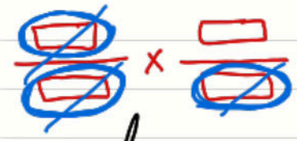
$r = \sqrt{\frac{\rho_e L}{\pi \times R}} \rightarrow \frac{r_1}{r_2} = \sqrt{\frac{\rho_{e1}}{\rho_{e2}} \times \frac{L_1}{L_2} \times \frac{R_2}{R_1}}$

$\lambda = \frac{h}{\sqrt{2m\,KE}} \rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{KE_2}{KE_1}}$

$h = 6.625 \times 10^{-34}\ J.s$

$m = 9.1 \times 10^{-31}\ Kg$

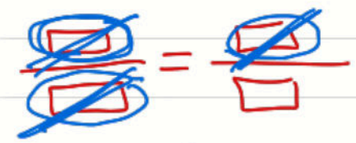
Simplify



example

$\frac{3\lambda^2}{5\lambda} \times \frac{3\lambda^2}{\lambda} = \frac{9\lambda^4}{5\lambda^2} = \frac{9}{5}\lambda^2$

$\frac{3\lambda}{5} \times \frac{3\lambda}{1} = \frac{9\lambda^2}{5}$



example

$\frac{3\lambda^2}{5\lambda} = \frac{3\lambda^2}{\lambda^3}$

$\frac{1}{5} = \frac{1}{\lambda^2} \rightarrow \lambda^2 = 5 \rightarrow \lambda = \pm\sqrt{5}$



Decimal Movement

$= 123.456 \times 10^{7}$

$= 12.3456 \times 10^{8}$

$= 1.23456 \times 10^{9}$

$= 0.123456 \times 10^{10}$

$= 1234.56 \times 10^{6}$

$= 12345.6 \times 10^{5}$

$\square . \square \times 10^{+} \quad \square . \square \times 10^{-}$

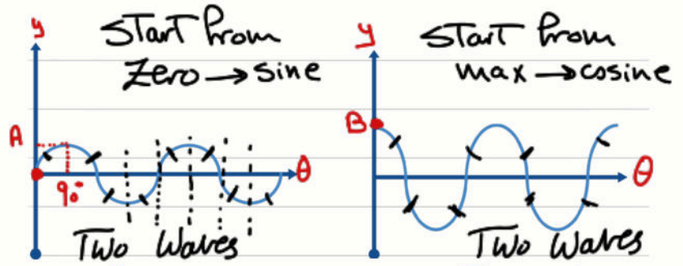
Example

$= 35.67 \times 10^{-8}\ m$

$= 356.7 \times 10^{-9}\ m$ ($nm \times 10^{-9} \rightarrow m$)

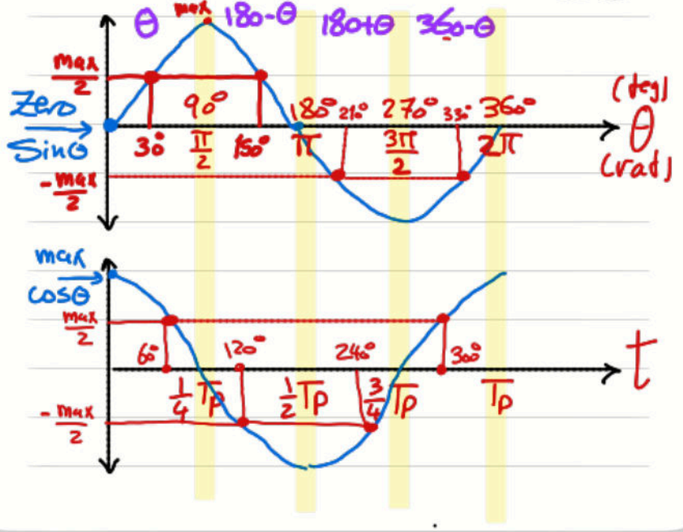
$= 356.7\ nm$

Periodic Time ← Waves → Frequency



$y = A \sin\theta$ $\quad y = B \cos\theta$

$$\text{Frequency} = \frac{\text{no. of Waves/cycles}}{\text{Time}} = \frac{1}{T_{per}}$$



Equations

One unKown ← Two unKowns → Three unKowns

1 One unKown

$$R = \frac{\rho L}{A} \rightarrow 10 = \frac{\rho_e \times 30}{20 \times 10^{-4}} \quad (\rho_e \rightarrow x)$$

shift Calc.

$$\rho_e = \frac{10 \times 20 \times 10^{-4}}{30}$$

2 Two unKowns

Note
⑤ equation
① $ax + by = c$

$I_1 + I_2 = 3$

$7I_1 - I_2 = 5$

$x \rightarrow I_1 = 1A$

$y \rightarrow I_2 = 2A$

	aI_1	bI_2	$c =$
1	1	1	3
2	7	-1	5

3 Three unKowns

Note
⑤ equations
② $ax + by + kz = d$

$I_1 + I_2 - I_3 = 0$

$6I_1 + 2I_2 = 10$

$5I_2 - 3I_3 = 1$

$x \rightarrow I_1 = 1A$

$y \rightarrow I_2 = 2A$

$z \rightarrow I_3 = 3A$

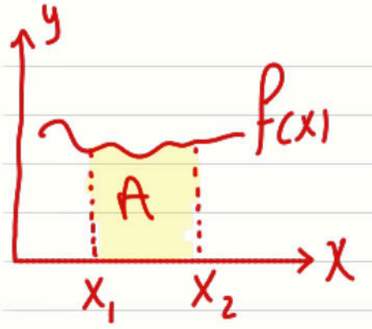
	aI_1	bI_2	cI_3	$d =$
1	1	1	-1	0
2	6	2	0	10
3	0	5	-3	1

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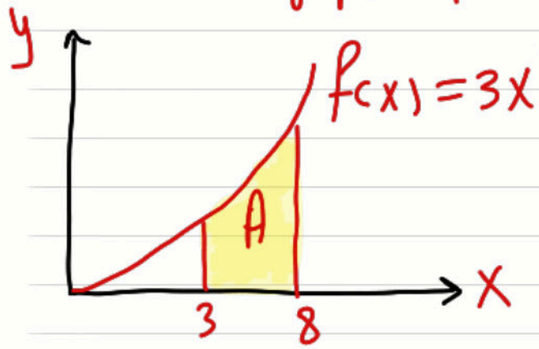


Integration

↳ Area Under The Curve

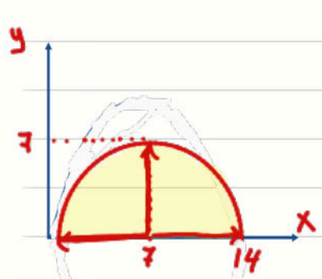


A = Area under the Curve/graph $= \int_{x_1}^{x_2} f(x)\,dx$

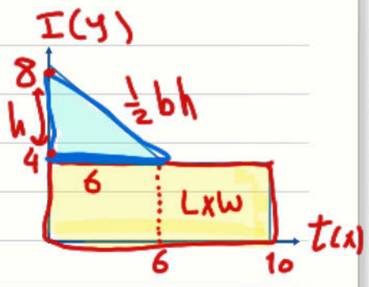


$$A = \int_3^8 3x^2 = 485$$

Area Under the Curve



M = y × X
= Area under the Curve/graph
$= \frac{1}{2}\pi r^2$
$= \frac{1}{2} \times \frac{22}{7} \times 7^2$
= 77



Q = I × t
= Area under graph
$= 10 \times 4 + \frac{1}{2} \times 6 \times 4$
= 40 + 12
= 52

A = The product of y-axis by X-axis

Notes

Calculas → Rate of Change

1. $f(x) = C \rightarrow f'(x) = \text{Zero}$
2. $f(x) = Cx \rightarrow f'(x) = C$
3. $f(x) = Cx^n \rightarrow f'(x) = ncx^{n-1}$
4. $f(x) = \sin(x) \rightarrow f'(x) = \cos(x)$
5. $f(x) = \cos(x) \rightarrow f'(x) = -\sin(x)$

$f(x) \xrightarrow{\frac{d}{dx}[\text{derv.}]} f'(x)$

$f(x) \xleftarrow{\int[\text{integration}]} f'(x)$

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